THE UNIVERSITY OF MELBOURNE

Student number

Assignment 1 Semester 1, 2023

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

Submission deadline: 5:00 PM (17:00 AEDT), Thursday 16 March 2023

This assignment consists of 5 pages (including this page)

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Instructions to Students

Writing

- If you have a printer (or can access one), print the assignment one-sided. Write your answers in the boxes provided on the assignment that you have printed.
- If you do not have a printer, and you know how to annotate a PDF using a tablet, then annotate your answers directly onto the assignment PDF and save a copy for submission.
- If you cannot use the template, write your answers on blank A4 paper. Mimic the template by numbering your pages, and note which answers to write on each page.
- Record your student ID number in the top right corner of this cover page (or the first page of your submission), and your details at the bottom of this page (or the first page).
- Provide complete and fully justified solutions for each question.
- Write legibly, and use appropriate notation. Marks will not be awarded to unclear work.
- If you run out of space for a question, append additional work to the end of your submission. Make a note ("See page X") in the relevant question box warning the marker.
- Teaching staff cannot help you directly with assignment questions. Please ask on the Ed Discussion board if you need clarification on the wording of a question.

Scanning

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Do not scan on an angle. Crop images to the page borders. Make sure that your PDF file is readable. Material that cannot be read will not be marked.

Submitting

- Go to the Gradescope window. Choose the relevant Canvas assignment. Make sure that you upload the correct PDF file. You should get a Gradescope confirmation email.
- Assignments submitted late will incur a penalty of 4% (of the total mark available) for every hour or part thereof up to 24 hours. Written assignments cannot be submitted more than one day (24 hours) late.

Student Details

- Student name: _____
- Tutorial A tutor name: _____
- Tutorial A day and time (e.g. Monday 9AM): _____

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Question 1

Suppose an association football (soccer) team consists of 11 players: one goal keeper, four defenders, four mid-fielders, and two strikers (forwards).

(a) There are try-outs for the football team where two people try-out to be a goalkeeper, seven try-out to be defenders, eight apply to be mid-fielders, and five apply to be strikers. How many different possible football teams could be formed?

(b) The association football team decides to make a futsal team. The futsal team will consist of five players from the original football team, and must include one goal keeper, at least one striker, and at least two defenders. How many different possible futsal teams could be formed from the original football team?

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Question 2

A small local school has 87 students in year seven. All year seven students must participate in at least one sport. Students can choose to participate in up to three sports. The sport options are: association football, basketball, and cross-country running.
Most students participate in only one sport:

- 24 students only play association football,
- 13 students only play basketball, and
- 37 students only participate in cross-country running.

Some students participate in two sports:

- 1 student plays both association football and basketball,
- 6 students participate in both association football and cross-country running, and
- 4 students participate in both basketball and cross-country running.

(a) How many year 7 students participate in all three sports?

(b) Show all the information on a Venn Diagram. You must define everything, and your solution must be large and easy to read.

(c) For each sport, each student gets a certificate of participation if they participated in that sport. In total, how many certificates need to be printed?

End of Assignment

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